CADEN DAVIS

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EDUCATION

Stanford University Los Angeles, CA
Master of Science in Electrical Engineering (MSEE) September 2025 – Graduating June 2027
Specialization in Signal Processing (DSP) and Wireless Communications | Breadth Areas in VLSI and ML

University of California, Los Angeles (UCLA) Los Angeles, CA
Bachelor of Science in Electrical Engineering (BSEE) September 2021 – June 2025
Cumulative GPA: **4.00**/4.00 | ECE Department Valedictorian 2025 | Fast Track to Success ECE Honors Program
Integrated Circuits and Systems Lab Alumni Scholar 2024 | Levi James Knight Jr. Scholar 2023 | Boeing Scholar 2022
GRE Score: V – 168/170, Q – 170/170, W – 5/6

EXPERIENCE

Cellular RF Firmware Intern San Diego, CA
Apple Inc. June 2025 – September 2025
- Point
- Point

DSP & Wireless Communications Intern Irvine, CA
Anduril Industries, Inc. June 2024 – September 2024
- Optimized detectors for frequency-hopping spread spectrum OFDM signals for 3x greater probability of signal intercept
- Wrote radio module firmware in C for LoRa chirp spread spectrum waveform design
- Parallelized DSP signal detection algorithms by using a polyphase filter bank to split bandwidth into multiple low-rate subchannels for parallel analysis with GPU of Nvidia Jetson embedded compute platform
- Developed RF system test platform to verify and study performance of signal detection and estimation algorithms

Wireless Research Assistant Los Angeles, CA
Wireless Lab, UCLA ECE September 2023 – Present
- Developed mmWave phased array control platform in Python as lab infrastructure for beamforming experiments
- Implemented range-Doppler radar for joint communication and sensing (JCAS) applied to next-generation traffic lights
- Created hands-on experiments with USRP and PlutoSDR software defined radios for lab outreach activities
- $1.5M project
- OFDM Comm based on 802.11a
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Electrical Hardware / Firmware Intern Irvine, CA
Anduril Industries, Inc. June 2023 – September 2023
- Designed schematic and PCB in Altium Designer for vehicle control user interface emphasizing design for test principles and robust on-board communication protocols including differential SPI / LVDS
- Developed firmware in C for user interface with low polling rate baremetal superloop prioritizing ease of future expansion

Networking Research Assistant Los Angeles, CA
Networked Embedded Systems Lab, UCLA ECE February 2022 – June 2023
- Simulated modulations and medium access control methods in MATLAB for narrowband simplex networks of underwater sensor nodes employing on-off keying (OOK) for acoustic communication
- Developed PCB and C++ interface for real-world sensor node prototype verified in Marina Del Rey harbor
- "LocoMote: AI-driven Sensor Tags for Fine-Grained Undersea Localization and Sensing," 2nd Author, *IEEE Sensors Journal*

Medical Device Engineering Intern Irvine, CA
Viseon Inc. June 2021 – September 2021
- Point
- Point

PROJECTS

Out-of-Order Superscalar RISC-V Processor | *Verilog, Intel Quartus Prime* September 2024 – December 2025
- Point

- Direct Mapped Cache
- Branchless, dataflow only
- Behavioral sim in C++

FPGA Audio Spectrum Visualizer | *SystemVerilog, Intel Quartus Prime* September 2023 – June 2024
- Wrote Cooley-Tukey FFT module with SystemVerilog on Intel FPGA for capturing spectra of microphone input signals
- Developed VGA interface and ping-pong RAM buffering of FFT outputs for smooth 60Hz display of spectrum to a monitor
- Expanded project with signal processing features including Hamming windowing, FIR filtering, and DC offset calibration

From-Scratch Superheterodyne Radio | *C, MATLAB, LTspice* September 2022 – June 2023
- Implemented BPSK signal processing pipeline in MATLAB including phase-synchronized coherent carrier demodulation with a digital PLL / Costas Loop and timing recovery using a Mueller-Muller sampling phase detector
- Rewrote application in C for real-time processing on STM32F4 microcontroller with STM32CubeIDE
- Designed and simulated amplifiers, mixers, oscillators, and filters for HF band Tx/Rx superheterodyne radio pair in LTspice

"Micromouse" Maze-Solving Robot | *C, Altium Designer, Autodesk EAGLE* September 2021 – June 2022
- Point
- Point

Japan-America Innovators of Medicine | *LTspice* June 2022 – July 2022
- Point
- Point

IoT Smart Sleepmask using Multicore-Enabled RTOS | *C, FreeRTOS* September 2024 – June 2025
- FreeRTOS dual-core, PlatformIO, ESP-IDF
- Core-crossing queue of structs for producer (sensors interfaced with I2C (HR, SpO2, Temp, Accel)) or raw DMA-mapped ADC (microphone))-consumer (MQTT networking)
- Second queue in producer (HTTP networking)-consumer (affectors interfaced with I2C (RGB LEDs) or raw DMA-mapped DAC (speaker))
- Digital FIR filtering of heart rate signal
- BPM algorithm based on zero-crossings of filtered HR signal

RC Human Project | *LTspice* May 2025
- Point
- Point

CAMPUS INVOLVEMENT

IEEE at UCLA, *Workshops Manager, Outreach Committee Representative*
ACM at UCLA - AI & Machine Learning Committee, *Technical Events Lead*
BMES at UCLA, *Sleepmask Project Firmware Lead*

COURSEWORK

In-Progress: Random Processes, Dynamical Systems, and VLSI Systems

Communications: Modern Wireless Communications I, II | RF Systems | Electromagnetics I, II
Signal Processing: Digital Signal Processing I, II | Audio & Image Processing | Computational Imaging | Applied Numerical Computing | Machine Learning & Neural Networks | Control Systems
VLSI: Digital Logic Design | Computer Architecture (RISC-V) | Digital Integrated Circuits & VLSI (Cadence Virtuoso) | Analog Integrated Circuits
Embedded: Data Structures & Algorithms (C++) | Computer Systems & Assembly (C, Git, Linux, GDB, OpenMP) | Operating Systems (POSIX)